JERES-J-901
Instrument Air Supply Systems

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0	JER Engineering	2008

Revision Tracking

Revision	Revision Date	Scope of Revision
0	29 April 2008	Issued for a bid package.
03	31 March 2008	Issued for bid package only for review.
02	17 December 2007	The first official JERES-J-901 is issued.
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00	05 February 2007	First draft issued is based on Saudi Aramco SAES-J-901 “Instrument Air Supply Systems” issue date 30 June, 2003. Document was modified to incorporate amendments and modifications.

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1 Scope

This Standard defines the minimum mandatory requirements governing the design and installation of instrument air supply systems.

2 Conflicts, Deviations and Clarifications

2.1 Any conflicts between this standard and other applicable Company Engineering Standards (JERES), Material Specifications (JERMS), Standard Drawings (JERSD), Engineering Procedures (JEREP), Company Forms or Industry standards, specifications, Codes and forms shall be brought to the attention of Company Representative by the Contractor for resolution.

Until the resolution is officially made by the Company Representative, the most stringent requirement shall govern.

2.2 Where a licensor specification is more stringent than those of this standard, the Licensor's specific requirement shall apply.

2.3 Where applicable Codes or Standards are not called by this standard or its requirements are not clear, it shall be brought to attention of Company Representative by Contractor for resolution.

2.4 Direct all requests for deviations or clarifications in writing to the Company or its Representative who shall follow internal Company procedure and provide final resolution.

3 References

The selection of material and equipment, and the design, construction, maintenance, and repair of equipment and facilities covered by this standard shall comply with the **latest edition** of the references listed below as of the CUT-OFF DATE as specified in the Contract unless otherwise noted.

3.1 Company References

Company Engineering Standards

JERES-A-112	*Meteorological and Seismic Design Criteria*
JERES-D-001	*Design Criteria of Pressure Vessels*
JERES-H-002	*Internal and External Coatings for Steel Pipelines and Piping*

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JERES-H-100	*Painting Requirements for Industrial Facilities*
JERES-H-101	*Approved Protective Coating Systems for Industrial Plant & Equipment*
JERES-J-003	*Instrumentation – Basic Design Criteria*
JERES-J-600	*Pressure Relief Devices*
JERES-J-601	*Emergency Shutdown Systems*
JERES-L-132	*Velocity Criteria and Basic Material Selection for Process Piping*
JERES-L-150	*Pressure Testing of Plant Piping and Pipelines*

Company Materials Specifications

JERMS-K-3102	*Packaged Plant and Instrument Air Compressors*
JERMS-J-4821	*Instrument Control Cabinets – Outdoor*
JERMS-J-4831	*Instrumentation for Packaged Units*

3.2 Industry Codes and Standards

Process Industry Practices (Reference only)

PCCIA001 *Instrument Air Systems Criteria*

4 Definitions

4.1 Definitions presented in this Standard and other JER documents have precedence over other definitions. Conflicts between various definitions shall be brought to the attention of Company Representative by the Contractor for resolution.

Company: Jubail Export Refinery.

Company Representative: A designated person from the Company or an assigned third party representative.

Company Inspector: A designated person or an agency responsible for conducting inspection activities on behalf of the Company.

4.2 List of Definitions

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5 Design

5.1 General

5.1.1 Instrument air supply systems shall provide dry, oil-free air to pneumatic instrumentation, valve actuators, and other services requiring instrument-quality air.

5.1.2 No cross-connections between plant or utility air services shall be made to the instrument air distribution system downstream of the air dryer inlet filters. A valve at the tie-in from the instrument air compressor(s) discharge to the utility air distribution system shall automatically close to isolate the two systems upon low instrument air system pressure.

5.1.3 The following pressure criteria shall be used to design the instrument air system.

a) Relief valve setting: Equipment MAWP

b) High pressure alarm: Between 125 psig and MAWP

c) Maximum normal operating pressure: 8,7b (125 psig)

d) Minimum normal operating pressure: 4.5b (65psig)

e) Start standby compressor(s): 4,9b (70 psig*)

f) Activate emergency shutdown: 4,2b (60 psig)

* *Typical. Actual value to be determined based on system design and capacity.*

5.1.4 All equipment of the air system shall be capable of being isolated by block valves.

5.2 Environmental Conditions

Unless otherwise specified, equipment will be installed outdoors in a desert area, with required relative humidity and temperature ratings as specified in JERES-J-003. Other criteria stated in JERES-J-003 shall also apply. Site specific meteorological and seismic data as specified in JERES-A-112 shall be used for equipment design.

5.3 Instrument Air Compressors

5.3.1 Compressors shall not use oil in any parts exposed to the compressed air (oil-free).

5.3.2 Centrifugal compressors are preferred for capacities above 17 standard cubic meters per minute (600 scfm).

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5.3.3 New air compressor systems shall be designed to provide always 100% backup capacity via one or more standby compressor(s). Two compressors, one main and the other standby, sized to individually provide 100% capacity are preferred. Compressors shall have both automatic and manual start capabilities. A manual switch shall be provided to permit selection of either of the compressors as the duty compressor.

5.3.4 When an existing instrument air system is being expanded, backup compression equal to the capacity of the largest primary system compressor shall be provided.

5.3.5 A check valve shall be provided between each compressor discharge and the first block valve.

5.3.6 Centrifugal Compressors

5.3.6.1 Centrifugal compressors, when specified, shall be supplied in accordance with JERMS-K-3102.

5.3.6.3 Centrifugal compressors shall be controlled to maintain constant discharge pressure. Multiple compressors operating in parallel shall be controlled to balance the load between them.

5.3.7 Screw Compressors

5.3.7.1 Screw compressors, when specified, shall be supplied in accordance with JERMS-K-3102.

5.3.7.2 Screw compressors shall operate continuously. Flow control, using a slide valve, shall load or unload the compressor as required.

5.3.8 The intake air filter(s) shall be located away from sources of dirt, moisture, and toxic, corrosive or flammable gases. The air intake filter(s) shall be installed at a height to avoid ground level dust and debris.

5.4 Instrument Air Receiver

5.4.1 An instrument air receiver shall be provided to damp out system fluctuations and to provide surge capacity in the event of duty compressor shutdown. The receiver should be sized to provide adequate time to permit startup of the standby compressor(s), as well as sufficient capacity to allow a safe shutdown. Typical residence time is 1.5 minutes of the rated compressor capacity.

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5.4.2 The air receiver shall be provided with an automatic and a manual drain for liquid removal.

5.4.3 The air receiver shall be designed and built in accordance with JERES-D-001, and be provided with a pressure gauge and safety relief valve.

5.4.4 The air receiver shall be internally prepared and coated in accordance with JERES-H-100 and JERES-H-101, APCS-2A.

5.5 Instrument Air Dryer

5.5.1 An instrument air dryer shall be supplied to deliver air at a maximum dew point of -40°C at system pressure, unless the process requires a lower dew point. In addition, the following inlet conditions shall be clearly stated to vendors for correct sizing of desiccant chambers: maximum temperature, maximum flowrate, maximum moisture saturation, and minimum and maximum pressures.

5.5.2 A regenerative desiccant dryer shall be used. Refrigeration dryers are not permitted.

5.5.3 The air dryer shall utilize two desiccant chambers, allowing one chamber to regenerate while the other chamber is in service. Regeneration shall consume less than 20% of the total air capacity. This consumption shall be included in the system air requirements.

5.5.4 An inline continuous moisture indicator shall be provided in the dryer discharge.

5.5.5 The desiccant shall be a type that does not disintegrate upon contact with water. Activated Alumina, SAMS stock number 26-126-418 or 26-126-465, is preferred.

5.6 Instrument Air Filters

5.6.1 A coalescing prefilter shall be provided upstream of the air dryer. The filter shall be capable of removing entrained droplets of oil or water, and dust or other foreign matter. Filters are to be fitted with automatic drains.

5.6.2 A 3-micron afterfilter shall be installed downstream of all desiccant dryers to prevent desiccant fines from entering the downstream system.

5.6.3 The pressure drop caused by the drying and filtering equipment when it is clean shall not exceed 35 kPa (5 psi) at the maximum design flowrate.

5.7 Instrumentation and Controls

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5.7.1 Compressor Controls

Instrument air compressors shall be furnished with the manufacturer's standard control system.

5.7.2 Standby Compressor Controls

5.7.2.1 The standby compressor shall start upon any one or more of the following events:

a) Failure of the duty compressor

b) Low pressure in the air receiver. A pressure transmitter is the preferred field instrument. The setpoint shall be in accordance with Paragraph 4.1.3.5

c) Operation of any one of the manual start switch(es)

5.7.2.2 Starting upon events "a" and "b" shall be automatic. Starting of the stand-by compressor shall not automatically stop the duty compressor(s).

5.7.3 Air Dryer Controls

5.7.3.1 Dryer controls shall be manufacturer's standard design, with preference given to programmable controller (PLC) systems.

5.7.3.2 The local control panel and all components contained therein shall meet the requirements of JERMS-J-4821, unless the air dryer is supplied as a package unit for which JERMS-J-4831 will apply.

5.7.3.3 Solenoid valves used for controlling dryer regeneration and duty cycles shall be capable of operating with air containing desiccant fines without sticking or clogging.

5.7.4 Instrument Air System Alarms

5.7.4.1 The following alarms shall be provided. For attended plants these alarms shall be in the appropriate control room. For unattended plants they shall be local with a high-visibility indication (flag).

a) High pressure in the instrument air receiver.

b) High liquid level in the instrument air receiver.

c) Shutdown of any operating (duty) compressor for any reason.

d) Starting of the stand-by compressor for any reason.

e) Change in selection of the stand-by compressor (change-over).

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f) Low air header pressure alarm.

5.7.4.2 Other alarms may be provided as engineering judgement dictates for various situations.

5.7.5 Plant Shutdown

5.7.5.2 The pressure transmitter generating the low-low instrument air pressure signal shall be installed per the requirements of JERES-J-601, downstream of all drying and filtering facilities.

5.7.6 Safety Relief Valves

5.7.6.1 Safety relief valves shall comply with JERES-J-600.

5.7.6.2 All safety relief valves for instrument air service shall be equipped with a lifting device.

5.7.6.3 Safety relief valves shall be installed on the air receiver and air dryer, and on any other equipment requiring overpressure protection.

5.8 Instrument Air Distribution System

5.8.1 General Requirements

5.8.1.1 Piping material shall be selected in accordance with JERES-L-132. The selection of alternative piping material requires the approval of the Company Representative.

5.8.1.2 All main and branch headers shall be sloped {minimum 64 mm per 30 m (2.5 inches per 100 feet)} and provided with low point drains.

5.8.1.3 All headers shall terminate with a line-size valve, blind, or plug to facilitate cleaning and plant expansions.

5.8.1.4 Each branch header shall connect to the top of the main header through a branch-line-size, full-bore isolation valve.

5.8.1.6 Supply takeoffs to individual instruments (a line serving up to a maximum of four devices) shall connect to the top of the branch header through a takeoff-line-size, full-bore isolation valve.

5.8.1.6.1 The minimum size of supply takeoffs is ½-inch.

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5.8.1.6.2 Twenty percent (20%) spare takeoff fittings and block valves shall be installed on the branch header.

5.8.1.7 Instrument air piping shall not be pocketed (i.e., shall not contain, U-sections). Instrument air piping shall not be installed underground.

Exception:

Instrument air branch connections may be buried to cross beneath roadways or fences where there is no existing overhead rack and where the aboveground support would interfere with crane or emergency vehicle movement. Prior approval from the Company Representative is required. Buried branch lines shall have low point drains, stainless steel construction and external coating per JERES-H-002.

5.8.2 Header and Takeoff Sizing

5.8.2.1 The pressure drop for piping between the dryer afterfilter outlet and the most remotely located user shall not exceed 35 kPa (5 psi) under maximum service flowrate.

5.8.2.2 For installations in which the instrument air consumers are located more than 600 m (2000 ft) from the instrument air compressor; pipe sizes should be calculated to keep pressure drops within the above limit. Below 600 m (2000 ft) instrument air header pipe size should be selected from Table 2.

Table 2

Number of Users	Pipe Size
1 - 4	½ inch
5 - 9	¾ inch
10 - 15	1 inch
16 - 80	1½ inch
81 - 150	2 inch
151 - 300	3 inch

5.9 Instrument Air System Capacity

5.9.1 Steady State Consumption

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5.9.1.1 The capacity of the instrument air supply system shall be based on the total requirements of all connected loads, assuming all instruments are in steady state operation.

Comment:

Company facilities typically take plant and utility air from the instrument air system upstream of the dryers. This intermittent demand is not considered when calculating steady state operation.

5.9.1.2 Consumption figures for each instrument should be obtained from the manufacturer's literature. When this is not available, default values from the Table 3 shall be used.

Default Air Consumption Values for Various Instruments

Table 3 - Air-Consumption Per Instrument

	m^3/hour (std)	**SCFM**
Control Panel Mounted Instruments (including pneumatic relays, etc.)	0.85	0.50
Field Mounted Instruments	0.85	0.50
Control Valve with Diaphragm Actuator and Positioner	1.275	0.75
Control Valve with Piston Actuator and Positioner	8.50	5.00

5.9.1.3 For capacity requirement purposes it shall be assumed that air-operated valves (AOVs) will operate once every eight hours and that when emergency shutdown valves (ZVs) operate, the instrument air system will be capable of returning the ZV storage tanks from 0 kPa (ga) to 700 kPa (ga) (100 psig) within one hour maximum. Shorter times may be specified when operating conditions so require.

5.9.1.4 In systems in which AOVs are designed to operate on a defined schedule (e.g., process dryer switching, or water deionization unit switching) their actual consumption shall be determined.

5.9.1.5 Purge consumption rates shall be the maximum flow of the purge rotameter.

5.9.2 Additional Capacity

5.9.2.1 To compensate for system leaks and allow for plant expansion, 20% of the steady state consumption shall be added.

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5.9.2.2 To account for air dryer regeneration usage, 20% of the sum of steady state consumption plus the additional capacity of Paragraph 5.9.2.1 shall be added.

5.9.2.3 For Screw instrument air compressors, 20% of the sum of steady state consumption plus additional capacities calculated in Paragraphs 5.9.2.1 and 5.9.2.2, shall be added to compensate for wear and loss of efficiency.

Note:

Centrifugal compressors do not require wear and efficiency compensation.

6 Testing

Pressure testing of supply system and distribution system per JERES-L-150, using dry instrument air or inert gas is required. Hydrostatic testing of instrument air piping is not permitted.

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